

Shiv Nadar University Chennai

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

Experiment 1

Implementation of a Single Layer Perceptron for Binary Classification

1. Objective

The objective of this experiment is to understand the workings of a single layer perceptron by implementing it from scratch for a binary classification task. The experiment involves reading the dataset, performing the necessary preprocessing steps, and training the perceptron using the perceptron learning algorithm. The experiment also focuses on observing how the perceptron's weights and bias change during training, analysing the effect of different learning rates on convergence, and evaluating the classifier using standard performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

2. Background Theory

An Artificial Neuron is the fundamental building block of a neural network. It receives one or more input features, multiplies them by corresponding weights, adds a bias term, and applies an activation function to determine the output.

The Single Layer Perceptron is the earliest supervised learning algorithm capable of solving linearly separable binary classification problems.

Mathematical Formulation

Weighted Sum

$$z = \mathbf{w}^T \mathbf{x} + b$$

where

- $\mathbf{x}$: Input vector
- $\mathbf{w}$: Weight vector
- b : Bias
- z : Linear combination

Prediction

$$\hat{y} = f(z)$$

Weight Update Rule

$$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta(y - \hat{y})\mathbf{x}$$

Bias Update

$$b_{new} = b_{old} + \eta(y - \hat{y})$$

where η denotes the learning rate.

3. Activation Functions

An activation function determines the output of a neuron based on the weighted sum of its inputs. It introduces non-linearity, enabling neural networks to learn complex patterns. Although the perceptron uses only the Step Activation Function, modern deep learning models employ differentiable activation functions for efficient optimization through backpropagation.

Step Activation Function

$$f(z) = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

The Step Activation Function produces a binary output and is therefore suitable only for linearly separable binary classification problems.

Activation	Output Range	Typical Applications
Step	$\{0, 1\}$	Classical Perceptron
Sigmoid	$(0, 1)$	Binary Classification Output Layer
Tanh	$(-1, 1)$	Hidden Layers (Traditional Networks)
ReLU	$[0, \infty)$	Hidden Layers in CNNs and MLPs
Leaky ReLU	$(-\infty, \infty)$	Avoiding Dying ReLU
Softmax	$(0, 1)$	Multi-class Classification Output Layer

Note: This experiment uses only the Step Activation Function. The remaining activation functions will be studied in subsequent experiments involving Multilayer Perceptrons and Deep Neural Networks.

4. Dataset Description

Dataset: Banknote Authentication Dataset

Source: UCI Machine Learning Repository

Download Link:

<https://archive.ics.uci.edu/dataset/267/banknote+authentication>

Instances	1372
Features	4 Numerical Features
Classes	2
Missing Values	None
Task	Binary Classification

Feature Description

- Variance
- Skewness
- Curtosis
- Entropy

Target Class

- 0 – Authentic Banknote
- 1 – Forged Banknote

5. Experimental Procedure

Task 1: Dataset Exploration

- Load the dataset.
- Display the first five samples.
- Determine dataset dimensions.
- Identify missing values.
- Display descriptive statistics.

Expected Output

Dataset summary and statistical information.

Task 2: Exploratory Data Analysis

Generate

- Histograms
- Correlation Heatmap
- Scatter Plot
- Boxplots

Task 3: Data Preprocessing

- Normalize all numerical features.
- Split the dataset into Training (80%) and Testing (20%).

Task 4: Perceptron Implementation

Implement from scratch

- Weight Initialization
- Bias Initialization
- Step Activation Function
- Forward Propagation
- Perceptron Learning Rule

Task 5: Model Training

Display

- Epoch Number
- Misclassified Samples
- Updated Weights
- Updated Bias

Task 6: Model Evaluation

Compute

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

1 6.Source Code

Listing 1: Python implementation of the Single Layer Perceptron

```

1 # -*- coding: utf-8 -*-
2 """Experiment1_Single_Layer_Perceptron.ipynb
3
4 Automatically generated by Colab.
5
6 Original file is located at
7     https://colab.research.google.com/drive/1nt0d4IDNtZm_PNrnxy13YRq9LAaSs102
8 """

```

```

9
10 # =====
11 # Experiment 1: Single Layer Perceptron
12 # Deep Learning Laboratory
13 #
14 # Environment Setup
15 # =====
16
17 !pip install -q pandas numpy matplotlib seaborn scikit-learn
18
19 # =====
20 # Import Libraries
21 # =====
22
23 import os
24 import random
25 import numpy as np
26 import pandas as pd
27
28 import matplotlib.pyplot as plt
29 import seaborn as sns
30
31 from pathlib import Path
32
33 from sklearn.model_selection import train_test_split
34 from sklearn.preprocessing import StandardScaler
35
36
37 print("Libraries imported successfully")
38
39 # =====
40 # Experiment Configuration
41 # =====
42
43
44 CONFIG = {
45
46     "dataset_path":
47     "data_banknote_authentication.csv",
48
49     "random_seed":
50     42,
51
52     "test_size":
53     0.20,
54
55     "plot_size":
56     (8,6),
57
58     "dpi":
59     300,
60
61     # Perceptron hyperparameters
62
63     "learning_rate": 0.01,
64
65     "epochs": 100
66
67

```

```

68 }
69
70
71 CONFIG
72
73 # =====
74 # Reproducibility
75 # =====
76
77
78 def set_seed(seed):
79     """
80     Fixes random seeds to ensure
81     repeatable experiments.
82     """
83
84     random.seed(seed)
85
86     np.random.seed(seed)
87
88
89
90 set_seed(CONFIG["random_seed"])
91
92 print("Random seed fixed")
93
94 from google.colab import files
95
96 uploaded = files.upload()
97
98 # =====
99 # Dataset Loading
100 # =====
101
102
103 COLUMN_NAMES = [
104     "variance",
105     "skewness",
106     "curtosis",
107     "entropy",
108     "class"
109 ]
110
111
112
113
114 def load_dataset(path):
115     """
116     Loads Banknote Authentication Dataset.
117
118     Parameters:
119     -----
120     path : str
121         Dataset location
122
123     Returns:
124     -----
125     DataFrame
126

```

```

127         Loaded dataset
128         """
129
130
131         dataframe = pd.read_csv(
132
133             path,
134
135             header=None,
136
137             names=COLUMN_NAMES
138
139         )
140
141
142         return dataframe
143
144
145
146 df = load_dataset(
147     CONFIG["dataset_path"]
148 )
149
150
151 df.head()
152
153 # =====
154 # Dataset Understanding
155 # =====
156
157
158 print("Dataset Shape:")
159 print(df.shape)
160
161
162 print("\nDataset Information:")
163 df.info()
164
165
166 print("\nMissing Values:")
167 print(df.isnull().sum())
168
169
170 print("\nStatistical Summary:")
171 df.describe()
172
173 plt.figure(figsize=CONFIG["plot_size"])
174
175
176 sns.countplot(
177
178     data=df,
179
180     x="class"
181
182 )
183
184
185 plt.title(

```

```

186     "Class Distribution"
187 )
188
189 plt.xlabel(
190     "Class"
191 )
192
193 plt.ylabel(
194     "Number of Samples"
195 )
196
197
198 plt.show()
199
200 # =====
201 # Feature Distribution
202 # =====
203
204
205 df.drop(
206     "class",
207     axis=1
208 ).hist(
209
210     figsize=(12,8),
211
212     bins=30
213 )
214
215
216
217 plt.suptitle(
218     "Feature Histograms"
219 )
220
221
222 plt.show()
223
224 plt.figure(figsize=(10,6))
225
226
227 sns.boxplot(
228
229     data=df.drop(
230         "class",
231         axis=1
232     )
233 )
234
235
236
237 plt.title(
238     "Feature Boxplots"
239 )
240
241
242 plt.xticks(rotation=45)
243
244

```

```

245 plt.show()
246
247 plt.figure(figsize=(8,6))
248
249
250 sns.heatmap(
251     df.corr(),
252     annot=True,
253     cmap="coolwarm",
254     fmt=".2f"
255 )
256
257 plt.title(
258     "Feature Correlation Heatmap"
259 )
260
261 plt.show()
262
263 plt.figure(figsize=(8,6))
264
265 sns.scatterplot(
266     data=df,
267     x="variance",
268     y="skewness",
269     hue="class",
270     palette="Set1"
271 )
272
273 plt.title(
274     "Class Distribution using Variance and Skewness"
275 )
276
277 plt.show()
278
279 # =====
280 # Save Generated Outputs
281 # =====
282
283 OUTPUT_DIR = Path("outputs")
284
285 OUTPUT_DIR.mkdir(
286     exist_ok=True
287 )

```

```

304 )
305
306
307 df.describe().to_csv(
308     OUTPUT_DIR / "dataset_statistics.csv"
309 )
310
311
312 print(
313     "Results saved successfully"
314 )
315
316 #Feature Separation
317 X = df.drop(
318     "class",
319     axis=1
320 )
321
322
323 y = df["class"]
324
325 #Train and Test split
326 X_train, X_test, y_train, y_test = train_test_split(
327     X,
328     y,
329     test_size=CONFIG["test_size"],
330     random_state=CONFIG["random_seed"],
331     stratify=y
332 )
333
334
335 #Normalization
336 scaler = StandardScaler()
337
338
339 X_train = scaler.fit_transform(
340     X_train
341 )
342
343
344 X_test = scaler.transform(
345     X_test
346 )
347
348
349 #Verification
350 print(X_train.shape)
351 print(X_test.shape)
352
353
354 #=====
355 # Additional Imports for Perceptron Implementation
356 #=====
357
358 import matplotlib.pyplot as plt
359
360

```

```

363 # =====
364 # Single Layer Perceptron Implementation
365 # =====
366
367
368 class SingleLayerPerceptron:
369
370     """
371     Implementation of a binary classifier
372     using the Perceptron learning algorithm.
373     """
374
375
376     def __init__(
377         self,
378         learning_rate=0.01,
379         epochs=100
380     ):
381
382         """
383         Initialize perceptron parameters.
384
385         Parameters
386         -----
387         learning_rate : float
388             Controls weight update magnitude.
389
390         epochs : int
391             Number of training iterations.
392         """
393
394
395         self.learning_rate = learning_rate
396
397         self.epochs = epochs
398
399         self.weights = None
400
401         self.bias = None
402
403         self.errors = []
404
405         self.weight_history = []
406
407         self.bias_history = []
408
409
410
411     def step_function(
412         self,
413         z
414     ):
415
416         """
417         Binary step activation function.
418         """
419
420
421         return np.where(

```

```

422         z >= 0,
423         1,
424         0
425     )
426
427
428
429     def predict(
430         self,
431         X
432     ):
433         """
434         Generate predictions.
435
436         Formula:
437
438          $z = Xw + b$ 
439
440         """
441
442         z = np.dot(
443             X,
444             self.weights
445         ) + self.bias
446
447         return self.step_function(z)
448
449
450
451
452
453
454
455
456     def fit(
457         self,
458         X,
459         y
460     ):
461         """
462         Train perceptron using
463         perceptron learning rule.
464         """
465
466
467         number_of_features = X.shape[1]
468
469
470         # Initialize parameters
471
472         self.weights = np.zeros(
473             number_of_features
474         )
475
476
477         self.bias = 0
478
479
480

```

```

481
482     # Training loop
483
484     for epoch in range(self.epochs):
485
486         misclassified = 0
487
488
489
490         for xi, target in zip(X, y):
491
492
493             prediction = self.predict(
494                 xi
495             )
496
497
498             update = (
499                 self.learning_rate
500                 *
501                 (target - prediction)
502             )
503
504
505             if update != 0:
506
507                 misclassified += 1
508
509
510
511                 # Update weights
512
513                 self.weights += (
514                     update * xi
515                 )
516
517
518                 # Update bias
519
520                 self.bias += update
521
522
523
524
525
526
527
528
529
530
531
532
533         # Store epoch information
534
535         self.errors.append(
536             misclassified
537         )
538
539

```

```

540         self.weight_history.append(
541             self.weights.copy()
542         )
543
544
545         self.bias_history.append(
546             self.bias
547         )
548
549
550
551         print(
552
553             f"Epoch {epoch+1}/{self.epochs} "
554             f"- Errors: {misclassified}"
555
556         )
557
558
559         return self
560
561
562 # =====
563 # Model Training
564 # =====
565
566
567 perceptron = SingleLayerPerceptron(
568
569     learning_rate=CONFIG["learning_rate"],
570
571     epochs=CONFIG["epochs"]
572
573 )
574
575
576
577 perceptron.fit(
578
579     X_train,
580
581     y_train.values
582
583 )
584
585 perceptron = SingleLayerPerceptron(
586
587     learning_rate=CONFIG["learning_rate"],
588
589     epochs=CONFIG["epochs"]
590
591 )
592
593
594 perceptron.fit(
595
596     X_train,
597
598     y_train.values

```

```

599 )
600 )
601
602 print("Final Weights:")
603 print(perceptron.weights)
604
605
606 print("\nFinal Bias:")
607 print(perceptron.bias)
608
609 plt.figure(
610     figsize=CONFIG["plot_size"],
611     dpi=CONFIG["dpi"]
612 )
613
614
615 plt.plot(
616     range(1, CONFIG["epochs"] + 1),
617     perceptron.errors,
618     marker="o"
619 )
620
621
622
623
624
625
626 plt.xlabel("Epoch")
627
628 plt.ylabel("Misclassified Samples")
629
630 plt.title(
631     "Training Error vs Epoch"
632 )
633
634
635 plt.grid(True)
636
637 plt.show()
638
639 predictions = perceptron.predict(
640     X_test
641 )
642
643
644 predictions[:10]
645
646 np.unique(
647     predictions,
648     return_counts=True
649 )
650
651 training_history = pd.DataFrame({
652     "Epoch":
653         range(1, CONFIG["epochs"] + 1),
654     "Errors":
655         perceptron.errors
656 }

```

```

658
659 })
660
661
662 training_history.head()
663
664 training_history.to_csv(
665     "perceptron_training_history.csv",
666     index=False
667 )
668
669 # =====
670 # Generate Predictions
671 # =====
672
673
674 y_pred = perceptron.predict(
675     X_test
676 )
677
678
679 print("First 10 predictions:")
680 print(y_pred[:10])
681
682
683 print("\nPrediction distribution:")
684 print(np.unique(
685     y_pred,
686     return_counts=True
687 ))
688
689 # =====
690 # Classification Metrics
691 # =====
692
693
694 from sklearn.metrics import (
695     accuracy_score,
696     precision_score,
697     recall_score,
698     f1_score
699 )
700
701
702
703 accuracy = accuracy_score(
704     y_test,
705     y_pred
706 )
707
708
709 precision = precision_score(
710     y_test,
711     y_pred
712 )
713
714
715 recall = recall_score(
716     y_test,

```

```

717     y_pred
718 )
719
720
721 f1 = f1_score(
722     y_test,
723     y_pred
724 )
725
726
727
728 metrics_result = pd.DataFrame({
729
730     "Metric":
731     [
732         "Accuracy",
733         "Precision",
734         "Recall",
735         "F1 Score"
736     ],
737
738     "Value":
739     [
740         accuracy,
741         precision,
742         recall,
743         f1
744     ]
745 })
746
747 })
748
749
750 metrics_result
751
752 # =====
753 # Confusion Matrix
754 # =====
755
756
757 from sklearn.metrics import confusion_matrix
758
759
760 cm = confusion_matrix(
761
762     y_test,
763
764     y_pred
765 )
766
767
768
769 plt.figure(
770
771     figsize=CONFIG["plot_size"],
772
773     dpi=CONFIG["dpi"]
774 )
775

```

```

776
777
778 sns.heatmap(
779     cm,
780     annot=True,
781     fmt="d",
782     cmap="Blues"
783 )
784
785 plt.xlabel(
786     "Predicted Label"
787 )
788
789 plt.ylabel(
790     "Actual Label"
791 )
792
793 plt.title(
794     "Confusion Matrix"
795 )
796
797 plt.show()
798
799 performance_table = pd.DataFrame({
800     "Dataset Size":
801     [len(df)],
802     "Train/Test Split":
803     ["80/20"],
804     "Learning Rate":
805     [CONFIG["learning_rate"]],
806     "Epochs":
807     [CONFIG["epochs"]],
808     "Accuracy":
809     [accuracy],
810     "Precision":
811     [precision],
812     "Recall":

```

```

835     [recall],
836
837     "F1-score":
838     [f1]
839
840 })
841
842
843
844 performance_table
845
846 len(perceptron.weight_history)
847 weights = np.array(
848     perceptron.weight_history
849 )
850
851
852
853
854 weights.shape
855
856 plt.figure(
857     figsize=CONFIG["plot_size"],
858     dpi=CONFIG["dpi"]
859 )
860
861
862
863
864
865 for i in range(weights.shape[1]):
866     plt.plot(
867         weights[:,i],
868         label=f"Weight {i+1}"
869     )
870
871
872
873
874
875
876 plt.xlabel(
877     "Epoch"
878 )
879
880
881 plt.ylabel(
882     "Weight Value"
883 )
884
885
886 plt.title(
887     "Weight Evolution During Training"
888 )
889
890
891 plt.legend()
892
893 plt.grid()

```

```
894 plt.show()
895
896 perceptron.bias_history
897
898
899 plt.figure(
900     figsize=CONFIG["plot_size"],
901     dpi=CONFIG["dpi"]
902 )
903
904 plt.plot(
905     perceptron.bias_history,
906     color="red"
907 )
908
909 plt.xlabel(
910     "Epoch"
911 )
912
913 plt.ylabel(
914     "Bias Value"
915 )
916
917 plt.title(
918     "Bias Evolution During Training"
919 )
920
921 plt.grid()
922
923 plt.show()
```

7. Mandatory Plots

Class Distribution

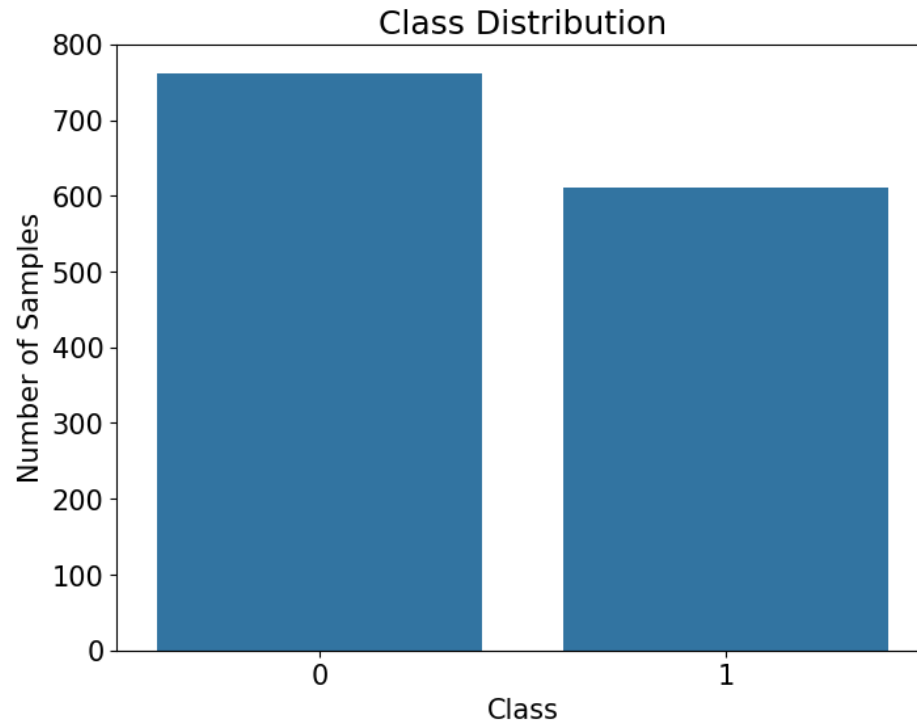


Figure 1: Class Distribution of the Banknote Authentication Dataset

The class distribution shows that both target classes are well represented in the dataset. Since the classes are reasonably balanced, the perceptron can learn without being significantly biased towards one class.

Feature Histograms

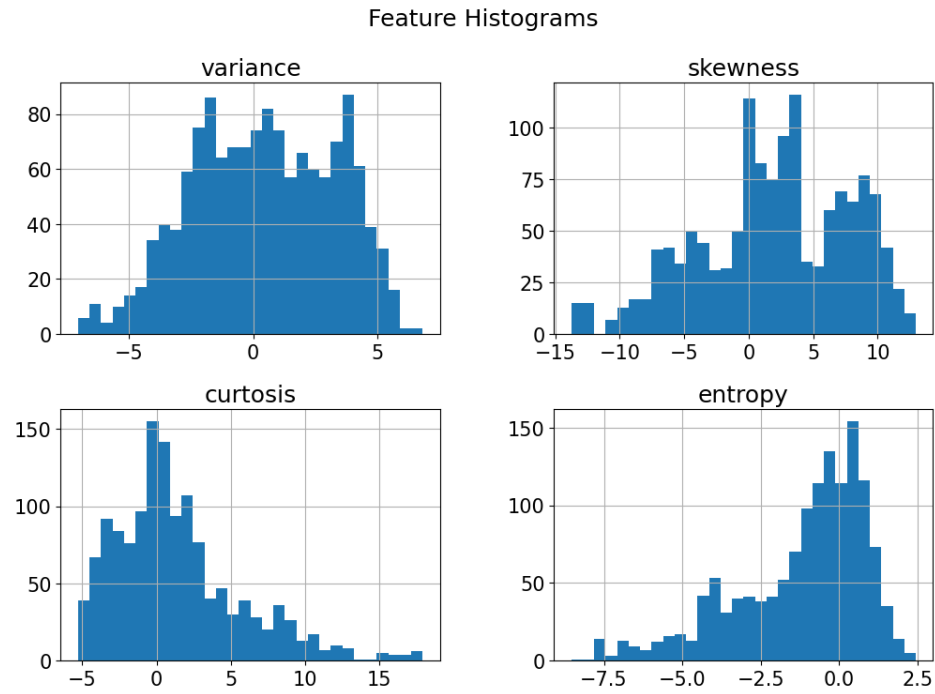


Figure 2: Distribution of Individual Features

The histograms illustrate the distribution of each numerical feature. The differences in their distributions indicate that each feature captures unique information that contributes to distinguishing authentic and forged banknotes.

Feature Boxplots

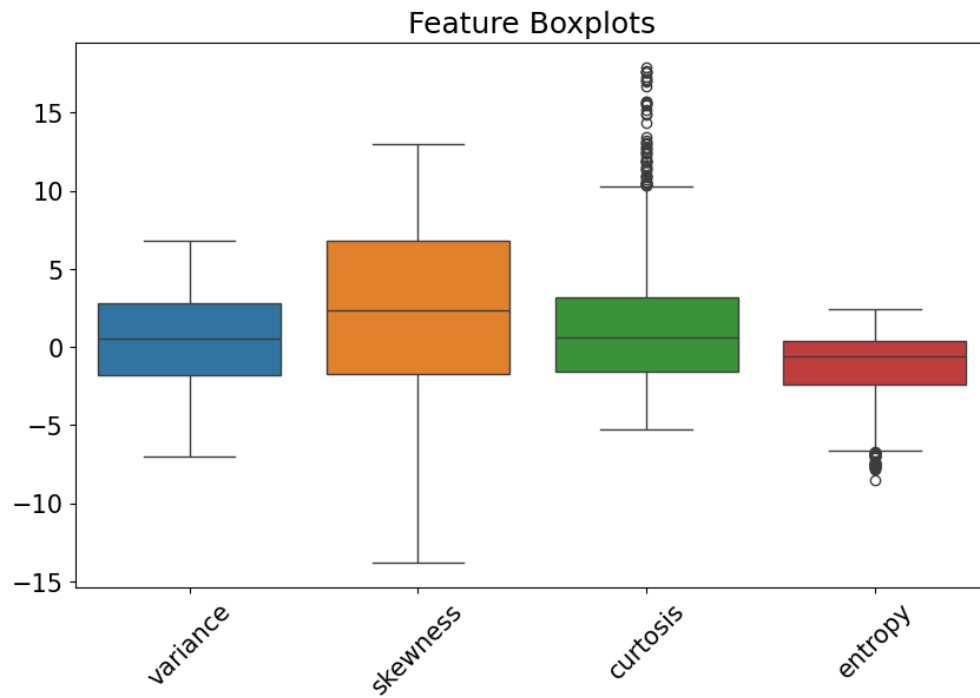


Figure 3: Boxplots of Input Features

The boxplots reveal the spread of the feature values and indicate the presence of several outliers. These observations represent natural variations within the dataset and were retained during model training.

Correlation Heatmap

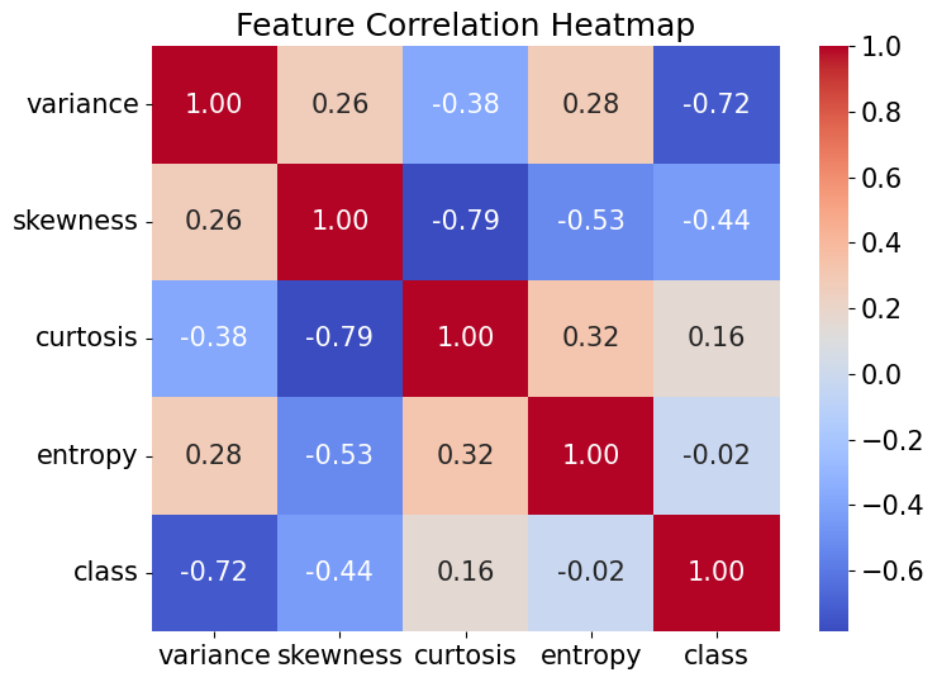


Figure 4: Feature Correlation Heatmap

The correlation heatmap highlights the relationships among the input features and the target class. Several features exhibit noticeable correlations, suggesting that they contain meaningful information for the classification task.

Scatter Plot

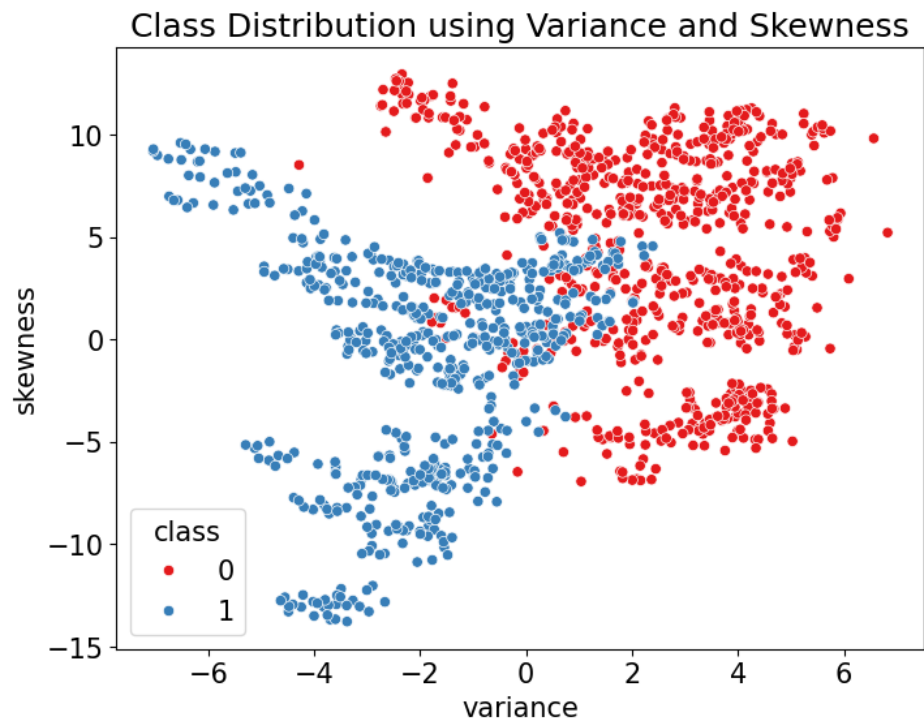


Figure 5: Scatter Plot of Variance versus Skewness

The scatter plot provides a visual representation of the two classes using the variance and skewness features. Although some overlap exists, a considerable amount of separation can be observed, making the dataset suitable for a Single Layer Perceptron.

Training Error

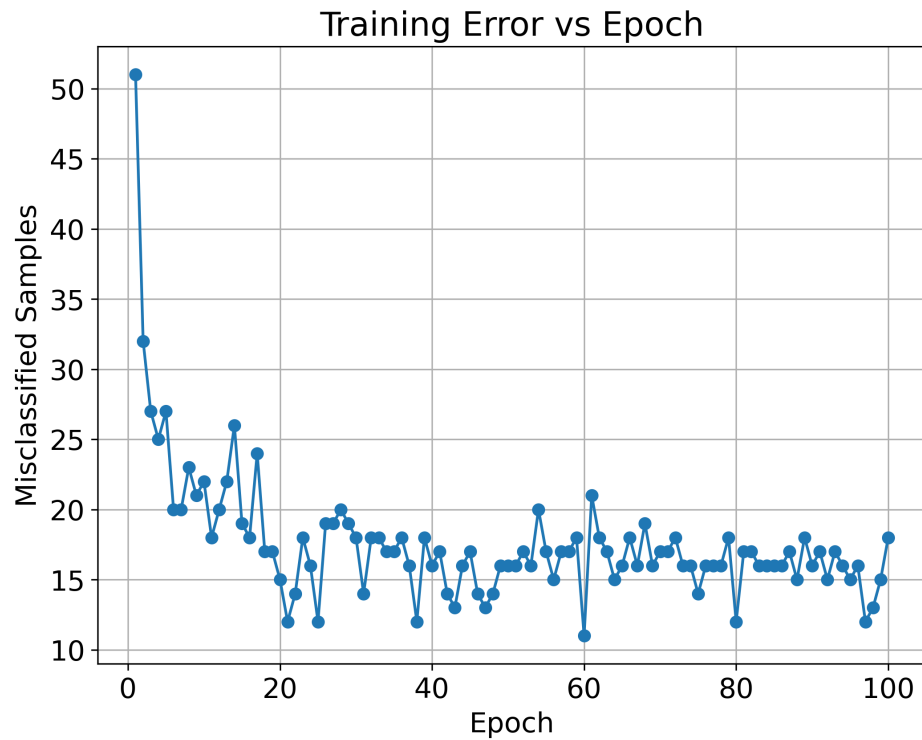


Figure 6: Training Error versus Epoch

The training error decreases significantly during the initial epochs and later stabilizes, indicating that the perceptron gradually learns and an effective decision boundary is set in place for the training data.

Weight and Bias Evolution during Training

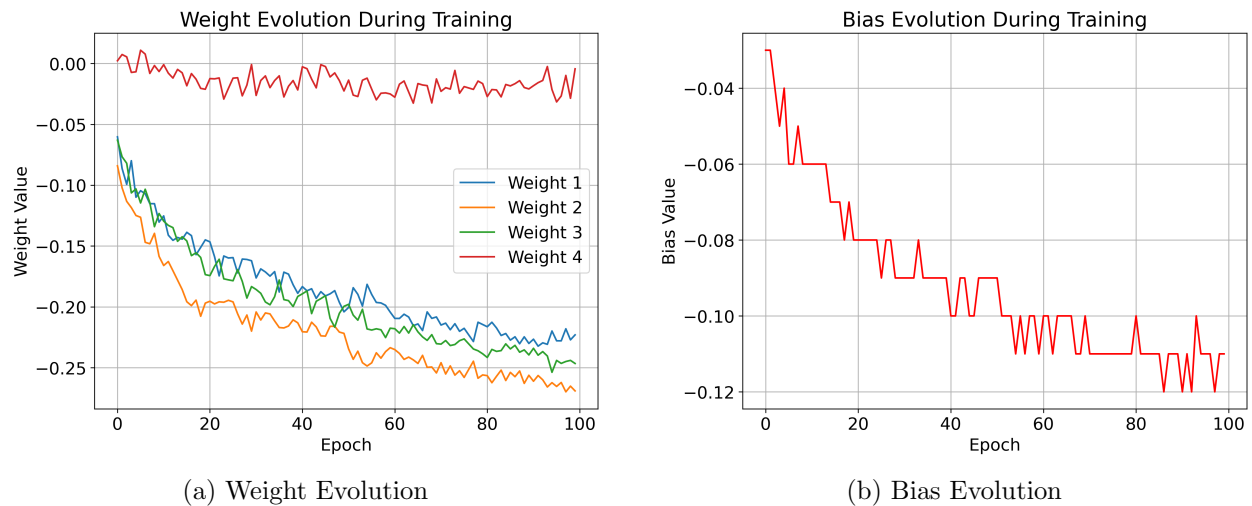


Figure 7: Evolution of the perceptron parameters during training.

The weights undergo relatively larger changes during the early stages of learning and gradually stabilize as training progresses, suggesting that the perceptron is approaching convergence.

A similar trend is observed for the bias term, indicating that the perceptron converges towards a stable solution as the number of training epochs increases.

Confusion Matrix

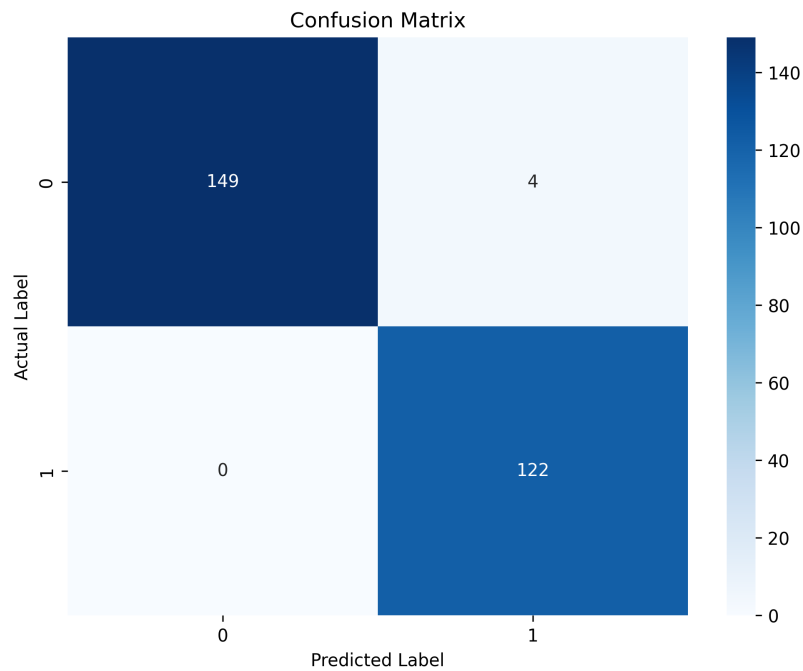


Figure 8: Confusion Matrix

The confusion matrix demonstrates that the majority of samples were classified correctly, with only a minuscule portion of misclassifications. This observation is consistent relative to the high accuracy, precision, recall, and F1-score obtained during evaluation.

8. Performance Tables

Training Summary

Dataset Size	1372
Train/Test Split	80/20
Learning Rate	0.01
Epochs	100
Final Weights	$[-0.22298399 \ -0.26903532 \ -0.24649669 \ -0.00438611]$
Final Bias	-0.10999999999999999
Accuracy	0.985455
Precision	0.968254
Recall	1.000000
F1-score	0.983871

Epoch-wise Learning

Epoch	Errors	Weight 1	Weight 2	Bias
1	51	...	...	...
...	...			
2	32	...	...	...
...	...			
3	27	...	...	...
...	...			
4	25	...	...	...
...	...			
5	27	...	...	...
...	...			
6	20	...	...	...
...	...			
7	20	...	...	...
...	...			
8	23	...	...	...
...	...			
9	21	...	...	...
...	...			
10	22	...	...	...
...	...			
⋮	⋮	⋮	⋮	⋮
⋮	⋮			
100	18	-0.222984	-0.269035	-0.246497
-0.004386	-0.110000			

9. Additional Tasks

A) Compare the Step and Sigmoid activation functions. Discuss why the Step Activation Function is not used in modern deep learning models

The step activation function generates only binary outputs, either 0 or 1. This depends on whether the input exceeds a predefined threshold or not, whereas, the sigmoid activation function generates a outputs between a continuous range spanning between 0 and 1, reminiscent of probability distribution.

One of the major flaws of step activation function is that it is non-differentiable and has a zero

gradient almost everywhere. Since modern deep learning models are trained using backpropagation and gradient-based optimization methods, the step function cannot be used effectively because gradients cannot be computed. The sigmoid function, differentiable in nature, allows gradients to propagate throughout the network, allowing optimization, although it may still be prone to the vanishing gradient problem.

B) Compare the implementation with Scikit-learn's Perceptron

The perceptron used in this experiment was built from scratch using NumPy. This implementation provides a thorough understanding of the perceptron learning rule, which involves manual initialization of weights and bias, prediction using step activation function, and repetitive parameter updates during training.

In comparison, Scikit-learn's `Perceptron` class is a superior optimization implementation that also provides regularization, early stopping, and parameter tuning. While Scikit-learn is more suitable for practical applications, manual implementation of the algorithm offers comprehensive analysis of the underlying learning process.

C) Repeat the experiment using learning rates 0.001, 0.01 and 0.1

When using a learning rate of 0.001, it resulted in slower convergence because the weight updates were very small. This required more epochs for the reduction of training error.

A learning rate of 0.01 provided stable learning and attained effective convergence within a specified number of epochs.

Increasing the learning rate to 0.1 caused larger weight updates, which initially accelerated learning but also resulted in fluctuations around optimal solution.

Among the three values, $\eta = 0.01$ provided the best balance between convergence speed and training stability for this dataset.

D) Explain why a Single Layer Perceptron cannot solve the XOR problem

Simply put, the XOR problem is not linearly separable. A Single Layer Perceptron can only learn linear decision boundaries using the perceptron learning rule. This implies that it cannot correctly classify all XOR input combinations regardless of the quantitative estimate of training epochs. Solving the XOR problem requires at least one hidden layer with nonlinear activation functions, which is what the network requires to learn complex nonlinear decision boundaries.

E) Study the effect of feature normalization on model convergence

Feature normalization scales all input features to a numerical range on a relative scale prior to training. Without normalization, features having larger numerical values dominate the weight updates, often resulting in slower or unstable convergence.

10. Experimental Results and Discussion

The single layer perceptron was implemented from scratch using NumPy and trained on the Banknote Authentication dataset, consisting of 1,372 samples. Before training, the dataset was normalised using the `StandardScaler` and divided into training and testing sets in an 80/20 split. The model was trained for 100 epochs($\eta = 0.01$)

During the initial stages of training, the data suggests that the perceptron had recorded 51 misclassified samples in the first epoch. As training progressed, the number of misclassifications dropped significantly, indicating that the model was gradually learning an effective decision boundary. Although minor fluctuations in the training error were observed throughout the later epochs, the learning process remained stable and converged with only 18 misclassified samples in the final epoch. The evolution of the weights and bias further confirmed the convergence of the learning algorithm, as the parameter updates became drastically smaller as the training progressed.

After the completion of the training, the perceptron converged to the final weight vector $[-0.222984, -0.269035, -0.246497, -0.004386]$ with a final bias value of $[-0.1100]$.

The trained classifier was then evaluated on the test dataset using standard performance metrics. The model achieved an accuracy of 98.55/100, hinting that almost all test samples were classified correctly. The precision of 96.83/100 indicates that the majority of banknotes predicted as forged were indeed forged, while the recall of 100/100 suggests that every forged banknote in the test set was successfully identified. Finally, the F1-score of 98.39/100 reflects a healthy balance between precision and recall, confirming the effectiveness of the classifier.

The confusion matrix showed that only a very small number of genuine banknotes were incorrectly classified, while all forged banknotes were correctly detected. Overall, the results obtained demonstrates that the implemented single layer perceptron is highly effective for the Banknote Authentication dataset.

11. Conclusion

This experiment provides a practical understanding of how a single layer perceptron performs binary classification, instead of relying on existing machine learning libraries. The perceptron, developed using NumPy, makes it possible to understand each stage of the learning process, from parameter initialization to weight updates using the perceptron learning rule.

The given banknote authentication dataset, after preprocessing and feature normalization, allows the model to be able to learn an effective decision boundary. The gradual reduction in training error, together with the stabilization of the weights and bias over successive epochs, gives conclusive evidence regarding the learning algorithm's ability of convergence. The evaluation metrics further confirms the effectiveness of the implementation, with the model achieving high accuracy, precision, recall, and F1-score on the test dataset.

All in all, the experiment successfully completes its objectives and builds a solid foundation for understanding more advanced neural network architectures and deep learning algorithms that build upon the principles explored in this implementation.

12. References

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